

LEARNING MADE EASY

Augury Special Edition

Machine Health

for
dummies[®]
A Wiley Brand



Reduce unplanned
downtime

Increase
production capacity

Improve asset
utilization

Compliments
of



AUGURY

Lawrence Miller

About Augury

Our mission is to provide manufacturers and other industrial sectors with insights into the health of machines, processes, and operations to transform how people work and what they can create. Together with our customers, we are building a world where the combined work of people and machines makes life better in every way. A pioneer in Machine Health and Process Health solutions, we use purpose-built AI, trained by industry experts and the world's largest data library, to help customers eliminate production downtime, improve process efficiency, maximize yield, and reduce waste and emissions. Augury's global customers achieve 3-10x ROI, often in a matter of months.

www.augury.com



Machine Health

Augury Special Edition

by Lawrence Miller

**for
dummies®**
A Wiley Brand

Machine Health For Dummies®, Augury Special Edition

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Introduction

Machines need regular maintenance to run properly, but figuring out what kind of maintenance is needed and how to prioritize those needs can be time-consuming and expensive endeavors. As a result, manufacturers frequently experience unplanned downtime events, which can lead to costly repairs, lost sales, worker injuries, and bad publicity.

Of course, downtime can't be completely avoided. No machine wears out or breaks down exactly on schedule. Planned downtime helps businesses avoid unplanned downtime by allowing teams to do scheduled maintenance, but too much planned downtime wastes scarce resources and eats into production time. It also prevents reliability teams from understanding what happens between maintenance periods. Manufacturers need a machine health solution to help them strike the right balance between maintenance and production.

About This Book

Machine Health For Dummies, Augury Special Edition, consists of six chapters that explore the following:

- » The basics of machine health: what it is, why it matters, what causes machine failures, and more (Chapter 1)
- » Different maintenance types, including reactive, preventive, and prescriptive (Chapter 2)
- » Important capabilities in a comprehensive machine health solution (Chapter 3)
- » Common machine health use cases (Chapter 4)
- » How to get started with machine health (Chapter 5)
- » Examples of machine health wins in manufacturing (Chapter 6)

Each chapter is written to stand on its own, so if you see a topic that piques your interest, feel free to jump ahead to that chapter. You can read this book in any order that suits you (though I don't recommend upside down or backward).

Foolish Assumptions

It's been said that most assumptions have outlived their usefulness, but I assume a few things nonetheless!

Mainly, I assume that you work in manufacturing, and you're looking for innovative solutions to proactively address machine health challenges, reduce downtime, and improve reliability. Perhaps you're a VP of manufacturing, engineering, or production. Or you may be a plant director, reliability leader, chief digital officer, chief data officer, IT manager, or maintenance manager.

If any of these assumptions describes you, then this is the book for you! If none of these assumptions describes you, keep reading anyway — it's a great book and after reading it, you'll have a healthy grasp of machine health.

Icons Used in This Book

Throughout this book, I occasionally use special icons to call attention to important information. Here's what to expect:



REMEMBER

This icon points out important information you should commit to your nonvolatile memory, your gray matter, or your noggin.



TECHNICAL
STUFF

If you seek to attain the seventh level of nerd-vana, then perk up! This icon explains the jargon beneath the jargon.



TIP

Tips are appreciated but never expected, and I sure hope you'll appreciate these useful nuggets of information.

Beyond the Book

There's only so much I can cover in this short book, so if you find yourself at the end of it wondering, "Where can I learn more?," head to www.augury.com.

- » Learning the basics of machine health
- » Making Industry 4.0 a reality with machine health
- » Examining the causes and impacts of machine failure

Chapter 1

Machine Health 101

In this chapter, you discover the *what* and *why* of machine health, how it enables Industry 4.0, and common causes of machine failures and their costs.

Machine Health: What It Is and Why It Matters

The arrival of the information age hasn't lessened our reliance on machines. We depend on them to produce essentials — such as power, food, and medicine — and to support nearly every aspect of our modern society. According to McKinsey and Company, machines drive 85 percent of gross domestic product (GDP) collectively in developed economies, and PwC predicts that 70 percent of GDP growth in the global economy will be driven by machines through 2030. When these machines fail or underperform, production lines and supply chains break. The National Association of Manufacturers estimates that nearly 10 percent of global productivity is lost due to unexpected machine failures. When a machine fails or a process is inefficient, production stops, waste goes up, worker safety is compromised, and energy is squandered.

Machine health brings the Internet of Things (IoT), artificial intelligence (AI), and reliability expertise together to predict and prevent equipment failure and improve overall machine performance. Full-stack machine health solutions are all-inclusive, from sensors and software to AI capabilities and human expertise, all working to eliminate machine failures. In the case of machine health for rotating equipment, which is the focus of this book, a full-stack solution includes sensors that capture vibration, temperature, and magnetic data from machines. Armed with this information, maintenance and reliability teams can leverage advanced AI diagnostics to improve the health and performance of industrial machines. Put simply, machine health empowers the people who design, maintain, operate, and manage industrial assets with complete visibility into the current health of each machine and how to improve it (see Figure 1-1).

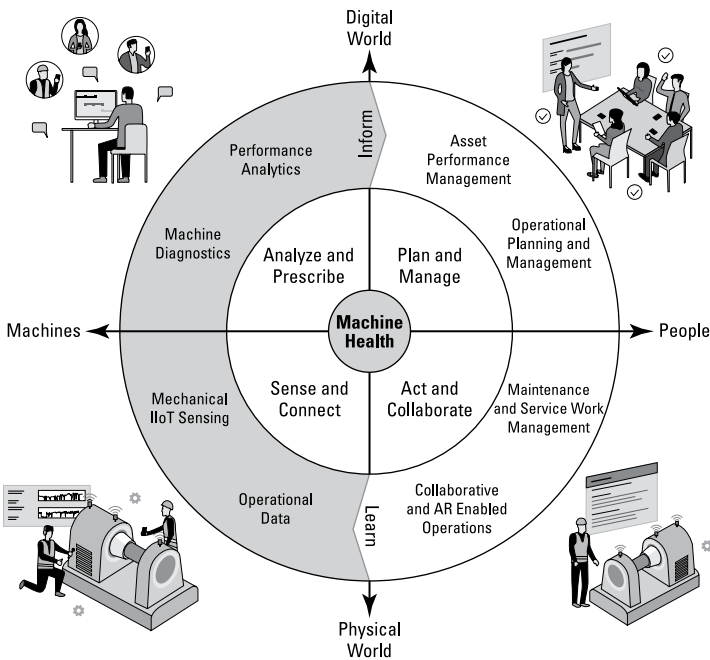


FIGURE 1-1: Machine health gives the people who design, maintain, operate, and manage industrial assets the power of real-time visibility and prescriptive insights.



TIP

Machine health helps manufacturers and industrial companies reduce production downtime, improve process efficiency, maximize yield, and achieve sustainability goals to realize the full potential of their production lines.

Machine Health and Industry 4.0

Before Industry 4.0, you didn't always know how a change in one aspect of production would impact everything else down the line. Computers were used extensively in manufacturing at the end of the twentieth century to produce relatively simple forecasts and analyze various if/then scenarios. However, applications were limited by computational power (now solved by cloud computing), the challenge of creating mathematical models and algorithms, and the inability of humans to find hidden combinations of factors that weren't obviously correlated.



TECHNICAL
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The Third Industrial Revolution introduced computing and communication technologies into manufacturing and production processes. The Fourth Industrial Revolution, or Industry 4.0, integrates automation and data exchange into manufacturing technologies and processes, leveraging innovations such as machine learning (ML), IoT, Industrial IoT (IIoT), and AI.

With Industry 4.0, analytics can deliver new insights by leveraging AI and other innovative technologies. For example:

- » **AI modeling:** AI can help you see whether a specific change to increase your throughput will negatively affect the quality of your product or increase energy usage to an unacceptable level.
- » **Cost reduction through predictive maintenance (PdM):** PdM leads to less maintenance activity, which means lower labor costs and reduced inventory and materials wastage.
- » **Predicting remaining useful life (RUL):** Knowing more about the behavior of machines and equipment leads to creating conditions that improve performance while maintaining machine health.

Machine health is a key enabler of Industry 4.0. It enables predictive and prescriptive maintenance by leveraging AI insights to

alert maintenance and reliability teams not only when machines will break down, but also what can be done to minimize mechanical degradation. The data and insights derived from machine health can improve cost and production predictions. In addition to routine maintenance, teams can perform preemptive repairs that lead to more consistent uptime and longer equipment life. As machines become more reliable, teams can also begin to continuously improve their regular maintenance practices, reducing unnecessary maintenance, further saving costs, and improving productivity. Additionally, better-running machines use less energy and produce less waste, moving manufacturers closer to their ideal environmental goals.

Machine health also enables greater collaboration between maintenance and reliability teams, leading to better decisions, improved operations, and new levels of expertise on the plant floor. Machine health insights provide leadership teams with a better idea of where to focus their attention when planning more innovative manufacturing initiatives. With machine health, machine reliability is no longer in question, thus reducing downtime, increasing capacity, improving maintenance spending, and creating safer work conditions.

Recognizing the Causes and Costs of Machine Failure

Machines fail for a variety of reasons, but the root causes can often be traced to a combination of the following factors:

- » **Poor data on machine conditions:** Even for companies that monitor their machines continuously, the lack of actual prescriptive insights on the health of machines means that maintenance teams spend hundreds of hours combing through data in search of a problem to solve, or worse, rely on archaic failure models that are mostly time or cycle based.
- » **No single source of truth:** There is no single digital source of truth for the health, efficiency, and performance of machines. Instead, there's a tangled web of maintenance, reliability, and operational systems and processes across the information technology (IT) and operational technology (OT)

domains, which are intended to keep machines running. Most companies need a more integrated, ecosystem approach to ensure that information is connected and insights are in context so they can make decisions based on complete and accurate information.

» **Siloed teams:** The health of a machine is directly related to its operating condition, but few industrial companies have a culture where maintenance, reliability, engineering, and operations teams collaborate and function as a single unit. Instead, these teams often have different objectives, report on different metrics, and make decisions based on separate data sets and incentives.

The costs of machine failure can add up quickly. According to Deloitte, industrial manufacturers lose \$50 billion annually due to unplanned downtime. Conversely, McKinsey and Company found that organizations with higher availability and a more efficient maintenance workforce realized a 4 percent to 10 percent increase in profitability.



REMEMBER

Machine health provides integrated, contextual information about the health of industrial machines, delivers timely insights, breaks down operational silos, democratizes institutional knowledge, and serves as a foundation for industrial companies to embrace the digital age.

IN THIS CHAPTER

- » Comparing maintenance and reliability
- » Recognizing different maintenance types
- » Advancing toward prescriptive maintenance
- » Leveraging artificial intelligence in machine health

Chapter 2

Current Approaches to Maintenance and Reliability

In this chapter, you learn how maintenance and reliability are related, the different types of maintenance and their associated challenges and benefits, how predictive and prescriptive maintenance benefits organizations, and the role of artificial intelligence (AI).

Maintenance and Reliability Basics

Maintenance and reliability are closely related but distinctly different. Maintenance focuses on performing maintenance during planned shutdowns or during unexpected shutdowns to fix failed equipment. Reliability is a strategic engineering task that focuses on understanding the causes of failures and preventing them from happening.

Different Maintenance Types

Maintenance programs typically fall into three categories: reactive, preventive, and predictive.

Reactive

The objective of reactive maintenance (also known as “run to fail”) is to repair an asset to its normal operating condition after poor performance is observed or a breakdown occurs. Changing the oil in your vehicle only when the “low oil pressure” indicator on your dashboard lights up (or when you see blue smoke coming out of your exhaust) would be an example of reactive maintenance.

The primary benefit of this approach is that, under normal operating conditions, reactive maintenance seems less costly than other types of maintenance because, until the machine breaks, you aren’t actually doing any maintenance. However, when — not if — machines break, reactive maintenance takes its toll: Emergency repairs typically cost three to nine times more than planned maintenance.

Preventive

Preventive maintenance, also known as time-based maintenance, requires maintenance tasks to be performed at set time intervals. Activities are planned and performed regardless of whether signs of deterioration are prevalent. Changing the oil in your car every 3,000 miles is an example of preventive maintenance.

The effectiveness of preventive maintenance is based on the assumption that the failure of assets can be predicted across time. Identifying the period that an asset is likely to operate without issues will lead you to ideal checkpoints to perform periodic maintenance.



TIP

There are some classes of assets for which preventive maintenance may be suitable, such as machines that are not necessary to keep production running but provide some important auxiliary function.

Predictive

Predictive maintenance consists of maintenance practices that monitor the performance and condition of equipment during normal operation to reduce the likelihood of failures. Also known as condition-based maintenance, manufacturers have used predictive maintenance since the 1990s. A smart car that collects analytics about your vehicle during operation and alerts you when specific maintenance tasks should be performed is an example of predictive maintenance.

The goal of predictive maintenance is to predict when equipment failure may occur (based on certain factors) and then prevent the failure through scheduled and corrective maintenance.



TECHNICAL
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Predictive maintenance cannot exist without *conditional monitoring*, which is the continuous monitoring of machines during production conditions to ensure the optimal use of those machines. There are three facets of conditional monitoring: online, periodic, and remote.

Evolving Toward Prescriptive Maintenance

Predictive maintenance — the current industry standard for monitoring and servicing machines — has created an entirely new challenge: too much data and no insights. That's why highly competitive manufacturers are moving beyond predictive maintenance to develop a prescriptive approach to maintenance and reliability.

With predictive (as well as reactive) maintenance, the root cause of a problem remains a mystery until a technician arrives to diagnose it. Prescriptive maintenance reduces both machine downtime and time spent diagnosing problems by using data analytics to identify what's compromising machine health and recommend the most effective course of action. Prescriptive maintenance is more targeted, actionable, and detailed than other types of maintenance — exactly what facilities need for more efficient and productive machines.

Discovering AI-Driven Machine Health

Machine health places the power of real-time visibility and prescriptive insights into the hands of the people who design, maintain, operate, and manage industrial assets. It creates a bedrock layer of information for the health and performance of industrial machines, breaking down operational silos, digitizing knowledge, and serving as a foundation for companies to meet new operational and business challenges. AI-driven machine health holds the promise of dramatic reductions in maintenance costs, labor costs, and inventory holding, while improving uptime, productivity, product quality, time to market, supply-chain resilience, risk reduction, and predictability.



TECHNICAL
STUFF

AI incorporates smart algorithms that are capable of interpreting information that typically require human intelligence and take hundreds of human hours to understand. Machine learning is a subset of AI algorithms, which learn by example and then apply what they've learned to new situations. The teaching and learning process takes time and must be ongoing if the algorithms are to improve in accuracy and capability. But algorithms alone are not enough. Any machine learning algorithm is only as accurate as the data that it learns from.

Here's an example: Machine health sensors continuously gather vibration, magnetic, temperature, and other types of data from machines. This anonymized data is then added to a training data set. Following a brief baseline period after the solution is first installed, reliability experts assign a machine health status based on the machine's condition. A healthy machine is classified as being in acceptable condition and doesn't require immediate attention, while a faulty machine is given a status of monitor, alarm, or danger. This provides a continuously growing pool of supervised examples to train the machine learning model by identifying data features indicative of machine health status.

To classify the health status of a new machine, data is extracted from the machine, faults are diagnosed, corrective actions are suggested, and the machine's health status is defined based on fault severity. Vibration analysts then validate the AI diagnosis, but with a significant scaling factor — instead of each analyst managing only a handful of machines, they can manage thousands.

IN THIS CHAPTER

- » Ensuring full machine coverage
- » Leveraging artificial intelligence
- » Empowering your employees
- » Offering a full suite of end-to-end services
- » Providing global coverage and scalability

Chapter 3

Key Capabilities for Machine Health

This chapter summarizes the key capabilities a machine health solution should have.

Comprehensive Asset Coverage

The last thing you need is another siloed solution on your shop floor. Your machine health solution should provide comprehensive coverage of all your rotating machine assets, including critical equipment, supporting equipment, and monitoring routes (see Figure 3-1).

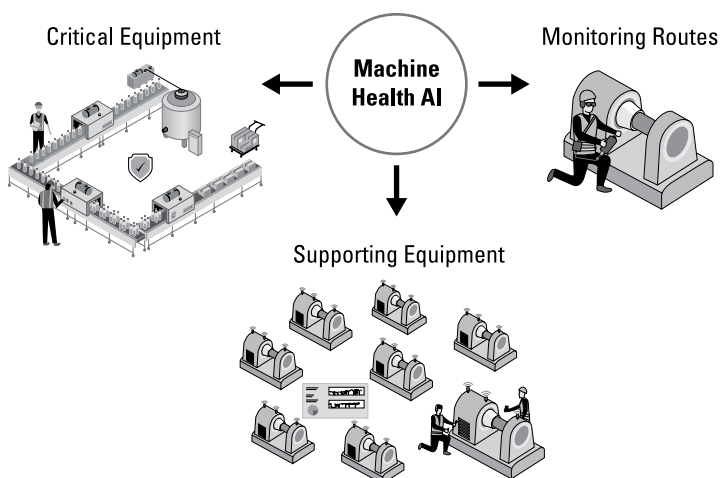


FIGURE 3-1: A complete machine health solution provides comprehensive asset coverage, including critical and supporting equipment, as well as monitoring routes.

Accurate, Prescriptive AI

Advanced artificial intelligence (AI) is used in machine health to continuously monitor and diagnose critical and supporting rotating equipment; the Internet of Things (IoT) and AI work together to predict and prevent machine failures and improve machine performance. By leveraging these technologies, machine health can elevate predictive maintenance programs, going beyond predicting machine failure by telling staff what to fix and how to fix it.



TIP

An AI-powered machine health solution can deliver outcomes that condition monitoring or simple threshold-based solutions could never achieve, including the following:

- » Diagnosing rotating machine issues early
- » Identifying malfunction type, root cause, severity, and the recommended course of remediation action
- » Regularly updating diagnostics for any number of machines
- » Continuously learning and improving from a growing data set
- » Ending firefighting and improving team productivity

True User Engagement and Autonomy

Advanced machine health technology doesn't replace workers on the ground. Instead, it enhances their work, enables more informed decision making, and allows them to bring their expertise to bear on many more machines without overloading their day. As leaders implement new technology for digital transformation in manufacturing, they must also focus on how that technology can empower their teams with valuable new skills.



TIP

Show workers how advanced tools can free up their time on routine tasks and create more capacity for problem solving, innovative thinking, and offering more value to the company than ever before. Think of AI as a copilot, not a replacement.

The people who actually touch the machines in your organization have the deepest understanding of them, so every technological implementation should be about empowering these individuals to deepen that understanding. Unlocking the full extent of their experience will ensure that the whole organization can benefit from it, allowing the power of machine health to scale far and wide.

End-to-End Services

Your machine health partner should provide more than just hardware and software. End-to-end services, including a dedicated account manager, as well as customer success, reliability, and 24/7 support teams can help you increase adoption across your organization and realize value at every stage of your machine health program. Professional services should include the following offerings:

- » Professional site assessments and installations
- » Asset strategy and maintenance/planned maintenance practice optimization
- » Prioritized deployment, rollout, and onboarding
- » Site-level goal attainment plan
- » Continuous support for analyzing and diagnosing machine problems and trends

- » Collaboration tools that engage with the organization and with solution experts to build knowledge capture and speed problem solving

Impact at Global Scale

Regardless of whether your organization is global today, your machine health program should be designed to meet the demands of growing organizations that require agility and scalability in the global economy. Hardware and electronic components should be approved to be deployed around the world and your machine health platform should support your team's preferred language.

IN THIS CHAPTER

- » Changing how you work
- » Looking at the cost of unplanned downtime
- » Maximizing production capacity
- » Getting more out of your assets
- » Achieving sustainability goals

Chapter 4

Machine Health and Business Value

In this chapter, you explore common machine health use cases and see how different organizations have used machine health to realize significant business value and benefits, such as greater collaboration, continuous improvement, environmental sustainability, and more.

Transform the Way You Work

In an era of labor shortages and technology innovation, manufacturing needs to move faster toward digitization. Using artificial intelligence (AI) in manufacturing, companies can eliminate repetitive tasks, reduce inefficiencies, and strengthen data-driven decision making.

DUPONT INNOVATES OPERATIONS USING MACHINE HEALTH

Committed to digital innovation, DuPont is using machine health to help transform their manufacturing operations. By moving from traditional, route-based maintenance to predictive maintenance, maintenance and reliability teams can address problems proactively, increasing productivity and avoiding unscheduled downtime. Jim Hunt, Corporate Digital Value Delivery Leader at DuPont, says, "In our implementations so far, we've had 100 percent accuracy in our predictions. We've had no misses."

In less than a year, DuPont has achieved 7x return on investment (ROI) at its pilot sites and continues to improve operations by rolling out machine health across all its businesses.

Learn more: www.augury.com/success-stories/duPont-sees-rapid-value-and-high-roi-with-predictive-maintenance?utm_campaign=mh_dummies.

When workers have insights into the real-time condition of their machines, they can break away from traditional maintenance schedules and manual tasks. Greater efficiency means more time for proactive work and collaboration with other departments, which leads to stronger cross-functional teams. It also leads to further operational innovations, such as process optimization.

Reduce Unnecessary Downtime

Unplanned downtime can be expensive. There's the cost of the repairs themselves, lost production and sales, and reputational damage. Maintenance and reliability teams must scramble to diagnose and fix the problem as quickly as possible — often in the middle of the night or over a weekend, meaning overtime pay. Sudden and catastrophic machine failures can jeopardize worker safety. And unplanned downtime can also depress worker morale by causing emotional stress. Thus, reducing unnecessary downtime has a significant impact on both direct and indirect costs.

ICL DIGITIZES MACHINE HEALTH TO REDUCE DOWNTIME

ICL is a leading global specialty minerals company that creates sustainable solutions for the food, agricultural, and industrial market.

ICL needed a digital solution that would provide actionable insights into their machine and reliability issues. Just ten months after implementing Augury Machine Health, they improved overall equipment effectiveness (OEE), upleveled their reliability workforce, increased asset life, and saved costs related to downtime, machine repairs, and production losses — all while driving digital transformation efforts forward. “Machine Health changed the way ICL approaches reliability by improving the visibility of our machines’ performance,” said Eyal Yaffe, Vice President of Operational Excellence & Innovation. “With Augury, ICL looks to continue being an innovator in the market, delivering value to customers around the globe.”

Learn more: www.augury.com/wp-content/uploads/Case-Study-ICL.pdf?utm_campaign=mh_dummies.

Increase Yield and Production Capacity

Industries that run continuous operations have limited scope to increase capacity. One short-term tactic to increase operating time is to reduce planned downtime. For example, instead of scheduling an 18-hour maintenance window every three weeks, a plant might instead take a few hours to do essential work, postponing major projects.

In industries that don’t have continuous operations — for example, a bottling plant with a five-day production schedule — employees may be asked to work overtime to make more product, or temporary employees may be brought in to add more hours to the schedule.

With real-time machine health insights, maintenance teams can adjust shutdown schedules based on current machine conditions, their history, and the recommendations of experts. With

OSEM-NESTLÉ TRANSFORMS OPERATIONS WITH DATA-DRIVEN INSIGHTS

Since implementing a machine health solution in 2020, food and beverage manufacturer Osem-Nestlé has greatly reduced unexpected downtime caused by machine failure.

This was recently demonstrated during an alert on Osem-Nestlé's Sabra hummus production line, which enabled teams to make a repair early, saving thousands of dollars and preventing a shutdown of the entire factory. With the hours saved, Osem-Nestlé has been able to increase capacity to better meet consumer demand.

Learn more: www.augury.com/success-stories/osem-nestle-transforming-operations-with-data?utm_campaign=mh_dummies.

these insights, they can increase output on production lines with healthier machines that are running optimally.

Improve Asset Utilization

Asset care is costly. A large portion of maintenance and repair budgets are related to acquiring and storing spare parts. Many manufacturers keep these in inventory because calendar-based maintenance dictates when parts should be replaced — even if replacement is not needed. They're trying to avoid the alternative: overspending up to five times the price, trying to get parts on short notice, and experiencing more downtime waiting for them to arrive. Whether spending money to hoard parts or paying more for last-minute parts, both methods are inefficient and harmful to a manufacturer's bottom line.

By optimizing these processes with machine health, manufacturers can save time and money, allowing resources to be allocated toward other aspects of production.

PAPER-TISSUE MANUFACTURER CUTS REPAIR AND REPLACEMENT COSTS, IMPROVES UPTIME WITH MACHINE HEALTH

Leaders at a multi-million-dollar manufacturer of personal health and hygiene products wanted to leverage a digital solution that could help them uplevel production processes, optimize maintenance and operations activities, and reduce machine downtime.

The benefits of their machine health solution were evident almost immediately, with some sites achieving overall ROI within six months. Over the long term, the company has avoided \$23 million in downtime and machine repair and replacement costs, making an average of 40 machine health improvements per month.

Learn more: https://info.augury.com/Paper-Tissue-Case-Study.html?utm_campaign=mh_dummies.

Reduce Loss, Waste, and Emissions

Being responsible for 30 percent of global emissions (some 16 billion tons of greenhouse gases), the manufacturing industry has become a lightning rod for government and media scrutiny around carbon emissions. This scrutiny has evolved into a real financial price tag. Manufacturers are increasingly being fined for the carbon dioxide they produce and incentivized to use greener alternative fuels and raw materials. Thus, reducing waste to improve sustainability and promote a healthier planet has become a top priority for manufacturers.

An industry study by the Electric Power Research Institute (EPRI) found that optimizing the performance of rotating assets — which account for approximately 54 percent of U.S. industrial electricity consumption — can reduce energy consumption by 12 percent to 15 percent. Thus, with machine health, better-performing machines can run at capacity and with less downtime, leading to less waste and more efficient energy use.

COLGATE-PALMOLIVE USES MACHINE HEALTH TO SAVE TIME, MONEY, AND ENERGY

Consumer packaged goods manufacturer Colgate-Palmolive Company and its brands are using machine health to digitally transform their manufacturing operations, improve efficiency, and meet global demand.

For example, machine health AI detected rising temperatures in the drive motor of a tube maker and alerted the plant team. They discovered a problem with the motor's water-cooling system and were able to prevent failure of the drive due to overheating. Early detection and repair saved 192 hours of downtime and 2.8 million tubes of toothpaste. Warren Pruitt, Vice President of Global Engineering Services at Colgate-Palmolive, said, "The benefits of having real-time access to machine health analytics have been so powerful, we're going to roll out this technology across our global supply chain."

Learn more: www.augury.com/success-stories/colgate-palmolive-optimizes-maintenance-using-ai-insights?utm_campaign=mh_dummies.

IN THIS CHAPTER

- » Choosing the right machine health solution
- » Prioritizing your machine assets based on business criticality
- » Leveraging deployment best practices
- » Getting a few quick wins
- » Looking ahead to production health

Chapter 5

Starting Your Journey with Machine Health

This chapter shows you how to get started with machine health to help ensure a successful deployment for your organization.

Identifying Your Machine Health Goals

Manufacturers need to address their biggest and most fundamental pain points first, instead of seeking out a solution that claims to solve all their problems. For example, if downtime is causing your plant to miss production targets and leading to worker frustration, start there. Then use the machine health insights you gain to build a foundation to address the next issue, such as improving overall maintenance costs or tackling quality and yield issues.

The machine health journey can happen in stages, starting with real, measurable problems. In this way, you can achieve a fast return on investment (ROI), rack up some quick wins to build upon, and create momentum leading to greater engagement across the organization.

Selecting a Vendor

With manufacturing firms increasingly adopting cloud-based software-as-a-service (SaaS) solutions, it's important to choose a vendor that does more than provide a new tool. You should evaluate their strategy around people, process, and technology.



TIP

When evaluating a vendor, consider the following:

- » **Partnership:** At every stage of your predictive maintenance journey, your vendor should provide strategic, technical, and product support to help your business grow and provide value at scale.
- » **Value:** Look for a fast time-to-value from baseline to production measured in days and weeks and return on investment (ROI) achieved in a matter of months.
- » **Expertise:** Select a vendor that has a diverse team with extensive manufacturing experience and a proven track record.
- » **Growth:** Ensure your partner is committed to the growth of your team with public and private user groups, discussion forums, certifications, and training.
- » **AI:** Be wary of vendors who overstate their AI's accuracy. Look for solutions that involve human feedback, because algorithms are enhanced by the users who use them.

Ensuring a Successful Deployment



TIP

Don't reinvent the wheel when it comes to your machine health journey. Here are some best practice tips based on the experiences of other industry professionals:

- » **Start with your team.** Assemble a team based on organizational structure, including members from safety, environmental, operations, and maintenance and reliability.
- » **Manage expectations.** Begin with clear and achievable goals and communicate changes and updates as necessary.

- » **Identify and train your champions.** They'll lead, educate, and increase the excitement and awareness of the program for the rest of the company.
- » **Educate the entire team.** Explain why you're collecting the data and how it will benefit each stakeholder. This creates more buy-in and less animosity — and a much greater likelihood of success.
- » **Realize machine health isn't a program; it's a culture.** Programs end. Machine health is a culture that can spread through an organization.
- » **Define success and celebrate wins.** It's essential to have a quantification plan for each asset and document the impact of any "wins" to help drive further engagement.
- » **Engage with the system.** Be proactive. Although machine health continuously monitors equipment, it can't replace the boots-on-the-ground perspective.

After Getting Your First Machine Health Wins

After you've achieved success in your first machine health roll-outs, take these steps to scale your initiative across your entire organization:

- » **Write a playbook.** A "lessons learned" playbook should outline best practices, forecast obstacles and challenges, and recommend workflows around using machine health data.
- » **Enlist influencers.** Influencers are people with hands-on experience who can accelerate adoption by working with technicians and managers at sites utilizing machine health data for the first time.
- » **Add new skills to expand value.** Machine health insights can uncover systemic machine problems at a facility. For example, a pattern of misalignment problems might indicate that it is worth investing in laser alignment training so fewer issues arise in the future.
- » **Derive new value.** While technology scales to new sites, let the existing sites evolve.

» **Expand your horizons.** Leverage interoperability to tie your machine health insights to your maintenance and reliability planning or with your operational information to provide new context into how production is impacting machines and vice versa.

Taking Your Next Steps to Production Health

By preventing machine failures and driving down maintenance and reliability costs, machine health provides the foundation for overall production health. The next step could involve process health, which provides teams with AI that can optimize production lines for quality and throughput while cutting waste and energy costs.

Production health, a fairly new concept in manufacturing, transforms industry by removing the friction created by competing business goals. Using AI-driven insights into machines, processes, and operations, companies can improve business outcomes, empower their workforce, and achieve sustainable production — all at the same time.



TIP

Learn more about production health's implications for manufacturing at www.augury.com/blog/production-health/production-health-is-the-next-leap-forward-for-manufacturing?utm_campaign=mh_dummies and www.augury.com/collateral/the-state-of-production-health-2023-augury?utm_campaign=mh_dummies.

Chapter 6

Ten (or So) Examples of Machine Health Wins

In this chapter, we share examples of machine health wins across different manufacturing industries:

- » **Paper products:** A leading manufacturer had ambitious goals for improving machine uptime, production capacity, and throughput while reducing carbon emissions. They installed machine health at 30 sites in 12 countries, resulting in more than \$23 million in estimated downtime avoidance and other cost savings, and achieved return on investment (ROI) within six months and time to value in 30 days at multiple sites.
- » **Building materials:** A leading building materials manufacturer needed to increase productivity and continue to digitally transform their operations. They deployed machine health to all 25 sites in 120 days, achieved ROI in less than 30 days, and implemented 260+ machine improvements in one year.
- » **Wood products:** Just four days after machine health was installed and a conveyor belt was placed in alarm, the maintenance team at a large wood products manufacturing plant replaced the motor, returning the machine to normal operation. Had this machine failed, the four-hour downtime disruption would have affected production at two other facilities, likely costing the company more than \$176,000.
- » **Oil and gas:** Machine health diagnosed a potentially catastrophic problem with a feed. A controlled shutdown and quick repair was initiated. During repairs, the on-site

maintenance team discovered molten nylon in the pump, which would quickly solidify and would have brought the machine to failure. The fast response helped avoid 252 hours of downtime and save \$4.5 million.

- » **Pet food:** A leading pet food manufacturer's compressor experienced sudden climbing vibration levels. Within 30 minutes, plant staff discovered a failing bearing and halted the compressor. The motor was replaced the next day, avoiding machine failure, which could have shut down two production lines for eight hours apiece, causing losses estimated at \$1,000 per minute for each line — a total of \$960,000.
- » **Chemicals:** A global manufacturer of agriculture, food, and engineered materials needed to digitally transform their machines and reliability programs. Within ten months of deploying machine health, they were able to achieve over \$1.1 million in savings associated with downtime costs, machine repairs, and production loss.
- » **Consumer packaged goods (CPG):** When machine monitoring detected increased vibration frequency in a CPG provider's extruder, the machine was stopped for inspection. The plant maintenance team immediately saw and replaced the faulty bearings. The machine was restarted with normal operation. A failure on this vital machine would've caused 480 hours of downtime.
- » **Pharmaceuticals:** The maintenance team at a large pharmaceutical manufacturing facility was alerted to faults on a packaging machine. The faults were discovered with AI technology within a machine health monitor. The maintenance team found extensive belt wear and replaced the motor, which returned the machine to normal operation. If not for the discovery, the company would've lost 2.1 million doses and experienced 6.5 hours of machine downtime.



Get insights into industry trends and best practices from the **Plant Services report**

2023 Machine Health Is Business Health

33% of manufacturers say their maintenance strategy includes more tech investment this year.

How is your company responding to asset management challenges, workforce shortages, and AI technology? See how your organization compares to industry peers in the "2023 Machine Health Is Business Health" report and find out:

- How the plant-floor and c-suite view the workforce skills gap
- Which maintenance approach ranks first for manufacturers
- How equipment failures are impacting production targets

Read the "2023 Machine Health Is Business Health" report now.



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Build a winning machine health culture

Machine health is vital to overall manufacturing and business health. The modern industrial world needs a holistic framework and category of solutions which helps people, processes, and systems to maximize machine productivity. This category — called machine health — transforms reliability, maintenance, operations, and asset performance management using artificial intelligence (AI) and Internet of Things (IoT) technologies, and will kickstart a new era of industrial reliability and performance.

Inside...

- Go beyond preventative maintenance
- Evolve to prescriptive maintenance
- Make Industry 4.0 a reality
- Transform how you work
- Improve production sustainability
- Leverage artificial intelligence



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